Stationary States and the Infinite Square Well

PHYS 2203 — Introductory Physics III

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Overview

Everything so far has described free particles. This lecture adds a potential energy $U(x)$ to the Schrödinger equation, so that it can describe particles that feel forces. Separating the equation into a spatial part and a time part leads to stationary states, whose probability density does not change in time, and to the time-independent Schrödinger equation. After stating the conditions a physically acceptable wave function must meet and reviewing classical bound states, we solve the equation for the simplest bound system, a particle in a box with infinitely high walls. Requiring the wave function to vanish at the walls allows only certain standing waves, and so only certain energies.

1 The Schrödinger Equation with a Potential

Everything so far, from the plane wave of Lecture 25 to the spreading wave packets of Lecture 28, has described **free particles**: no forces act, and all of the energy is kinetic. Most particles of interest are not free. An electron in an atom is held by the electric attraction of the nucleus, and an electron in a metal or a semiconductor feels forces from the surrounding atoms. To describe such particles, we must include the forces in the Schrödinger equation.

For a free particle (Lecture 25), the plane wave $Ae^{i(kx-\omega t)}$ solved the Schrödinger equation provided

$$\frac{\hbar^2 k^2}{2m} = \hbar\omega \qquad \Longleftrightarrow \qquad \frac{p^2}{2m} = E \qquad \Longleftrightarrow \qquad \text{KE} = E. \tag{1}$$

The free-particle equation is an accounting of energy in which all the energy is kinetic. To include a conservative force, we add its potential energy $U(x)$:

$$(\text{KE} + U)\,\Psi = E\,\Psi. \tag{2}$$

This suggests the **time-dependent Schrödinger equation**:

$$\boxed{-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + U(x)\,\Psi(x,t) = i\hbar\,\frac{\partial\Psi(x,t)}{\partial t}} \tag{3}$$

2 Stationary States

Separation of variables. Look for solutions that are a product of a spatial part and a time part:

$$\Psi(x,t) = \psi(x)\,\phi(t). \tag{4}$$

Substituting into the Schrödinger equation and dividing both sides by $\psi(x)\phi(t)$,

$$\underbrace{-\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2}+U(x)}_{\text{depends only on } x} = \underbrace{i\hbar\,\frac{1}{\phi(t)}\frac{\mathrm{d}\phi(t)}{\mathrm{d}t}}_{\text{depends only on } t}\,. \tag{5}$$

The left side cannot change when t changes, and the right side cannot change when x changes. Since x and t vary independently, both sides must equal the same constant C, the **separation constant**. (This requires U to depend on x only, not on t.)

The time part.

$$i\hbar\,\frac{\mathrm{d}\phi}{\mathrm{d}t} = C\phi \qquad \Longrightarrow \qquad \phi(t) = e^{-i(C/\hbar)t} = \cos\!\left(\frac{C}{\hbar}t\right) - i\sin\!\left(\frac{C}{\hbar}t\right). \tag{6}$$

This oscillates with the single angular frequency $\omega = C/\hbar$. A single frequency means a single, well-defined energy, $E = \hbar\omega$, so the separation constant is the energy:

$$C = E, \qquad \boxed{\Psi(x,t) = \psi(x)\,e^{-i(E/\hbar)t}} \tag{7}$$

The probability density does not depend on time.

$$|\Psi(x,t)|^2 = \left[\psi^*(x)e^{+i(E/\hbar)t}\right]\left[\psi(x)e^{-i(E/\hbar)t}\right] = |\psi(x)|^2. \tag{8}$$

Nothing observable about the particle changes with time, so these solutions are called **stationary states**.

This resolves a classical puzzle. An electron orbiting a nucleus is an accelerating charge, so classically it should radiate and spiral inward, and atoms should be unstable. In a stationary state the electron is a "cloud" whose charge density does not change, so it does not radiate.

The space part. With $C = E$, the spatial half of the equation becomes the **time-independent Schrödinger equation**:

Time-independent Schrödinger equation.

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + U(x)\,\psi(x) = E\,\psi(x)$$

The time dependence is always $e^{-i(E/\hbar)t}$, so from now on we work with $\psi(x)$. The potential energy $U(x)$ is what distinguishes one problem from another, just as the net force does classically.

3 Well-Behaved Wave Functions

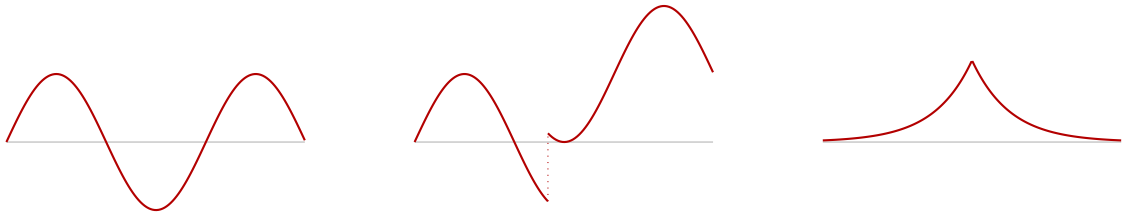
Not every mathematical solution is physically acceptable. A wave function must also be:

1. Normalizable. The total probability must be 1 (Lecture 29), so ψ cannot become infinite, and it must fall to zero as $x \to \pm\infty$.

2. Smooth.

* $\psi(x)$ **must be continuous.** A sudden jump acts like a wave of zero wavelength, meaning infinite momentum and kinetic energy.

* $\mathrm{d}\psi/\mathrm{d}x$ **must be continuous,** so that $\mathrm{d}^2\psi/\mathrm{d}x^2$, and so the kinetic energy, is finite. The one exception is at a point where $U(x)$ jumps to infinity, such as an infinitely hard wall.



(a) smooth: allowed

(b) ψ jumps: not allowed

(c) $\mathrm{d}\psi/\mathrm{d}x$ jumps (a kink): only where $U \to \infty$

Figure 1: (a) A smooth wave function. (b) A jump in ψ acts like a wave of zero wavelength, so it would mean infinite momentum and kinetic energy. (c) A kink, where ψ is continuous but its slope changes suddenly, makes $\mathrm{d}^2\psi/\mathrm{d}x^2$ infinite there. It is allowed only at a point where $U(x)$ is infinite, such as an infinitely hard wall.

4 Classical Bound States

A **bound state** is one in which a force restricts a particle to a finite region of space. Classically, total energy $E = \mathrm{KE} + U$ is conserved.

Mass on a spring. $U = \frac{1}{2}\kappa x^2$ is a parabola, and E is a horizontal line (Figure 2). The kinetic energy is the height of the E line above $U(x)$. At the **turning points** $x = \pm A$, KE $= 0$ and the mass turns around. Beyond them KE would be negative, so this is the **classically forbidden region**. For any E, the mass is bound.

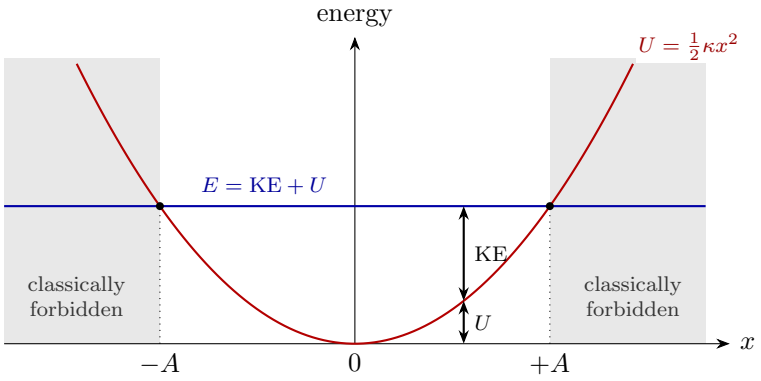


Figure 2: Energy versus position for a mass on a spring. The particle is confined between the turning points $\pm A$. The shaded regions are classically forbidden.

Quantum bound states. Looking ahead, quantum bound states share some features with classical ones and differ in others.

* Same: the total energy is constant, and is shared between kinetic and potential energy.

* Different: instead of a position at each time, we find a wave function. Bound states are **standing waves**, and just as a string with fixed ends allows only certain frequencies, only certain *discrete energies* are allowed.

* Different: the wave function can extend beyond the classical turning points, so there is a probability of finding the particle where classically it cannot be.

This lecture and the next two solve the time-independent Schrödinger equation for three simple potentials: the infinite well, the finite well, and the harmonic oscillator.

5 The Infinite Well

A particle is confined between two impenetrable walls a distance L apart. Inside, it is free to move.

$$U(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & x < 0 \text{ or } x > L \end{cases} \tag{9}$$

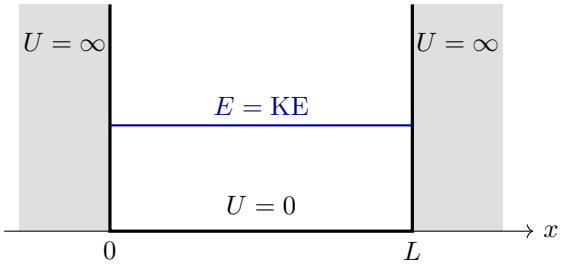


Figure 3: The infinite well. The particle is free between the walls and cannot get past them.

Outside the walls the particle can never be found, so $\psi(x) = 0$.

Inside the well, $U = 0$, and the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = E\psi \qquad \Longrightarrow \qquad \boxed{\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = -k^2\psi, \qquad k \equiv \frac{\sqrt{2mE}}{\hbar}} \tag{10}$$

The second derivative is a negative constant times the function itself, just like a mass on a spring, $\mathrm{d}^2x/\mathrm{d}t^2 = -(\kappa/m)x$. The solutions are sinusoidal. The general solution is

$$\psi(x) = A\sin kx + B\cos kx. \tag{11}$$

6 Boundary Conditions and Quantized Energies

The wave function must be continuous, and it is zero outside, so it must be zero at both walls.

Left wall, $x = 0$**:** $\psi(0) = A\sin 0 + B\cos 0 = B = 0$. So $\psi(x) = A\sin kx$.

Right wall, $x = L$: $\psi(L) = A \sin kL = 0$. We cannot set $A = 0$ (there would be no particle), so

$$\boxed{kL = n\pi, \qquad n = 1, 2, 3, \ldots} \tag{12}$$

This innocent-looking condition means the energy is **quantized**. Using $k = \sqrt{2mE}/\hbar$,

$$\frac{\sqrt{2mE}}{\hbar} L = n\pi \qquad \Longrightarrow \qquad \boxed{E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots} \tag{13}$$

The particle can have only these energies, in the ratios $1 : 4 : 9 : 16 : \cdots$

Same result from standing waves. A standing wave fixed at both ends has $\lambda_n = 2L/n$. With $p = h/\lambda$,

$$E = \frac{p^2}{2m} = \frac{h^2}{2m\lambda_n^2} = \frac{h^2 n^2}{8mL^2} = \frac{n^2\pi^2\hbar^2}{2mL^2}. \tag{14}$$

Quantum-mechanical bound states are standing waves.

Normalization. The total probability must be 1:

$$\int_0^L A^2 \sin^2\!\Big(\frac{n\pi x}{L}\Big)\, \mathrm{d}x = A^2\, \frac{L}{2} = 1 \qquad \Longrightarrow \qquad A = \sqrt{\frac{2}{L}}. \tag{15}$$

Infinite well.

$$\psi_n(x) = \begin{cases} \sqrt{\dfrac{2}{L}} \sin \dfrac{n\pi x}{L}, & 0 < x < L \\ 0, & \text{otherwise} \end{cases} \qquad\qquad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \ldots$$